

Title: Theory and Modeling of Frequency Response in RC
Network

Course: Circuit Theory

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Introduction

In electrical engineering, the frequency response describes how a system reacts to sinusoidal inputs of different frequencies. Understanding this behavior is essential in applications such as signal processing, filtering, and communications systems.

This project focuses on analyzing how system parameters influence the frequency response of an RC network, particularly in terms of the magnitude attenuation and phase shift of the output signal.

The RC network serves as a fundamental circuit in electrical engineering, often utilized for its simplicity and effectiveness in filtering and signal conditioning. By examining the relationship between resistance and capacitance, we can gain insights into how changes in these components affect the overall performance of the system. Such analysis not only aids in designing more efficient circuits but also provides a deeper understanding of the underlying principles governing dynamic systems. Through this study, practical implications for real - world applications will be explored, emphasizing the importance of precise parameter selection to achieve desired outcomes.

Theoretical Analysis

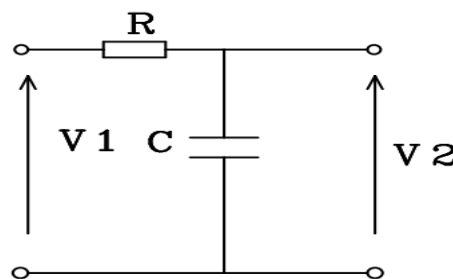


Figure 1. Link diagram of the proposed RC circuit

To make the calculation process more intuitive, the circuit can be represented as follows.

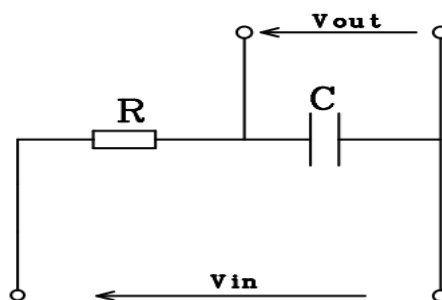


Figure 2. Intuitive Circuit Diagram of RC Circuit

Analysis of RC Low-pass Filter from Time Domain to Frequency Domain

In time-domain analysis, the capacitor is described by the differential equation:

$$i(t) = C * \frac{dv(t)}{dt}$$

This formulation is not convenient for steady

- state sinusoidal analysis, therefore frequency
- domain representation is introduced.

Impedance Representation

In the frequency domain, circuit elements are represented using impedance:

Resistor:

$$Z_R = R$$

Capacitor:

$$Z_C = \frac{1}{j\omega C}$$

This transformation simplifies the analysis by converting differential equations into algebraic expressions.

Derivation of Transfer Function

The transfer function is defined as the ratio between output and input in the frequency domain:

$$T(j\omega) = \frac{V_{out}(j\omega)}{V_{in}(j\omega)}$$

Voltage Divider Principle

The circuit can be treated as a voltage divider consisting of a resistor and a capacitor. The output voltage is taken across the capacitor:

$$V_{out} = V_{in} \times \left(\frac{Z_C}{Z_R + Z_C} \right)$$

Substituting the impedance expressions into the voltage divider formula yields the transfer function:

$$T(j\omega) = (1 / j\omega C) / (R + 1/(j\omega C))$$

To eliminate the fraction in the denominator, multiply both the numerator and denominator:
:

$$\frac{1}{j\omega C} \times j\omega C = 1$$

$$j\omega C \times \left(R + \frac{1}{j\omega C} \right) = 1 + jR\omega C$$

$$T(j\omega) = \frac{1}{1 + jR\omega C}$$

Magnitude Response

Mathematical Derivation

For complex numbers:

$$z = a + jb$$

Its modulus is:

$$|z| = \sqrt{a^2 + b^2}$$

Apply to denominator:

$$|1 + jR\omega C| = \sqrt{1 + (R\omega C)^2}$$

Therefore:

$$|T(j\omega)| = \frac{1}{\sqrt{1 + (R\omega C)^2}}$$

This represents the magnitude of the transfer function. As the frequency changes, the value of $|T(\omega)|$ will vary accordingly. When ω approaches zero, the magnitude approaches its maximum value of 1. As ω increases, the magnitude decreases, demonstrating the low-pass filter characteristic of the circuit. The rate of decrease is determined by the values of R and C, which together define the circuit's time constant. This mathematical relationship is crucial for analyzing how the circuit responds to different input frequencies.

Summary of its properties

The system response depends on input frequency.

At low frequency, the capacitor behaves like an open circuit, so the output voltage is nearly equal to the input. ($\omega \rightarrow 0$, $|T| \rightarrow 1$)

At high frequency, the capacitor behaves like a short circuit, causing the signal to be bypassed to ground. ($\omega \rightarrow \infty$, $|T| \rightarrow 0$)

Phase Response

Mathematical Derivation

complex phase:

$$\theta = \arctan\left(\frac{b}{a}\right)$$

denominator phase:

$$\theta = \arctan(R\omega C)$$

Due to:

$$\text{angle}\left(\frac{1}{z}\right) = -\text{angle}(z)$$

obtained:

$$\varphi(\omega) = -\arctan(R\omega C)$$

Physical Meaning

The negative phase indicates that the output signal lags behind the input signal.

$$\omega \rightarrow 0 \rightarrow \varphi \approx 0^\circ$$

$$\omega \rightarrow \infty \rightarrow \varphi \approx -90^\circ$$

Cut-off Frequency

Definition

The cut-off frequency is defined as the frequency at which the decreases to $1/\sqrt{2}$ of its maximum

value:

$$|T| = \frac{1}{\sqrt{2}}, (\sqrt{2} \approx 0.707)$$

Derivation

$$\frac{1}{\sqrt{(1 + (R\omega_0 C)^2)}} = \frac{1}{\sqrt{2}}$$

Square:

$$\begin{aligned} 1 + (R\omega_0 C)^2 &= 2 \\ (R\omega_0 C)^2 &= 1 \end{aligned}$$

Obtained:

$$\begin{aligned} R\omega_0 C &= 1 \\ \omega_0 &= \frac{1}{RC} \end{aligned}$$

Increasing R or C shifts the cut

- off frequency to lower values, making the system more selective to low
- frequency signals.

$$f_0 = \frac{1}{2\pi RC}$$

The cut-off frequency represents the boundary between pass-band and stop-band, where the output power drops to half of its maximum value.

Engineering Meaning

Use the dB formula to enhance the description

$$L = 20\log_{10} \left| \frac{1}{\sqrt{(1 + (R\omega C)^2)}} \right|$$

can be derived from the numbers:

$$20\log_{10}|0.707| \approx -3\text{dB}$$

The magnitude is often expressed in decibels (dB) to better visualize system behavior over a wide frequency range.

When the value of $R\omega C$ is significantly greater than 1, the original formula can be expressed as:

$$L = -20\log_{10}(\omega RC)$$

Through the analysis of formulas 5 and 6 (When the cutoff frequency is reached):

Amplitude = -3 dB

Phase = -45°

The system begins to exhibit significant attenuation

System Characteristics

The circuit represents a first-order linear time-invariant (LTI) system. At high frequencies, the magnitude behaves approximately as $1/(\omega RC)$, which results in a slope of -20 dB per decade. This characteristic confirms that the system is first-order.

Simulation (Python)

The implementation was carried out in Python. The complete source code is available in the accompanying GitHub repository.

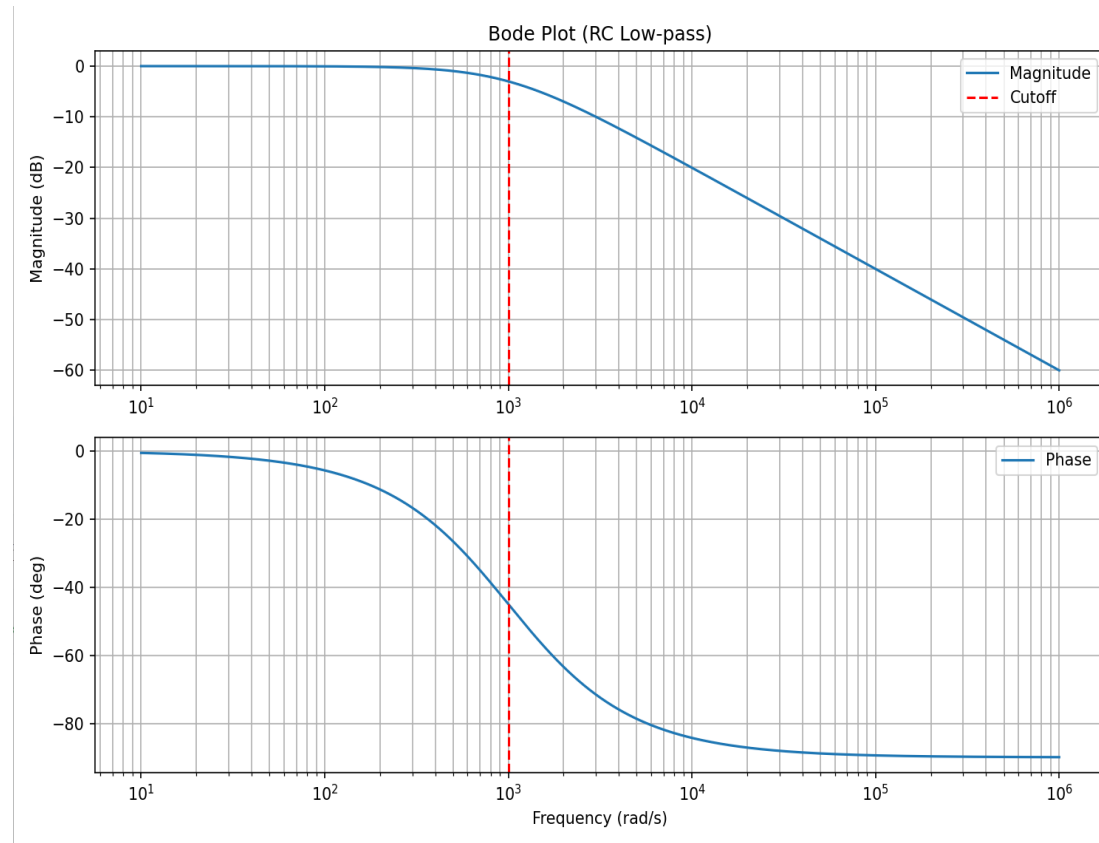


Figure 3. Bode plot of the RC low-pass filter

The magnitude response remains close to 0 dB at low frequencies, indicating that low-frequency signals pass through the system with minimal attenuation. As the frequency increases, the magnitude decreases, showing that high-frequency components are increasingly attenuated.

The cutoff frequency, marked by the dashed vertical line, corresponds to the point where the magnitude drops to -3 dB. At this frequency, the phase shift is approximately -45° , which is consistent with the theoretical behavior of a first-order system.

The slope of the magnitude response beyond the cutoff frequency is approximately -20 dB per decade, confirming that the circuit behaves as a first-order low-pass filter.

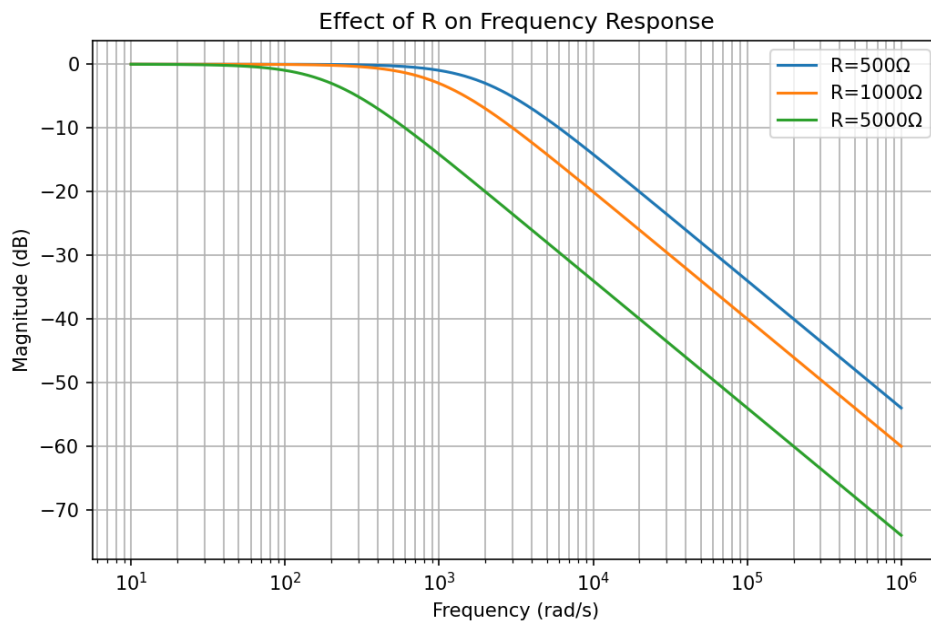


Figure 4. Effect of resistance on the frequency response of the RC circuit

The figure shows that increasing the resistance shifts the cutoff frequency toward lower values, causing the transition from pass-band to stop-band to occur at lower frequencies.

For smaller resistance values, the cutoff frequency is higher, allowing a wider range of frequencies to pass through the system. Conversely, larger resistance values result in stronger attenuation of higher-frequency signals.

This behavior is consistent with the relationship $\omega_o = 1/(RC)$, where increasing R reduces the cutoff frequency.

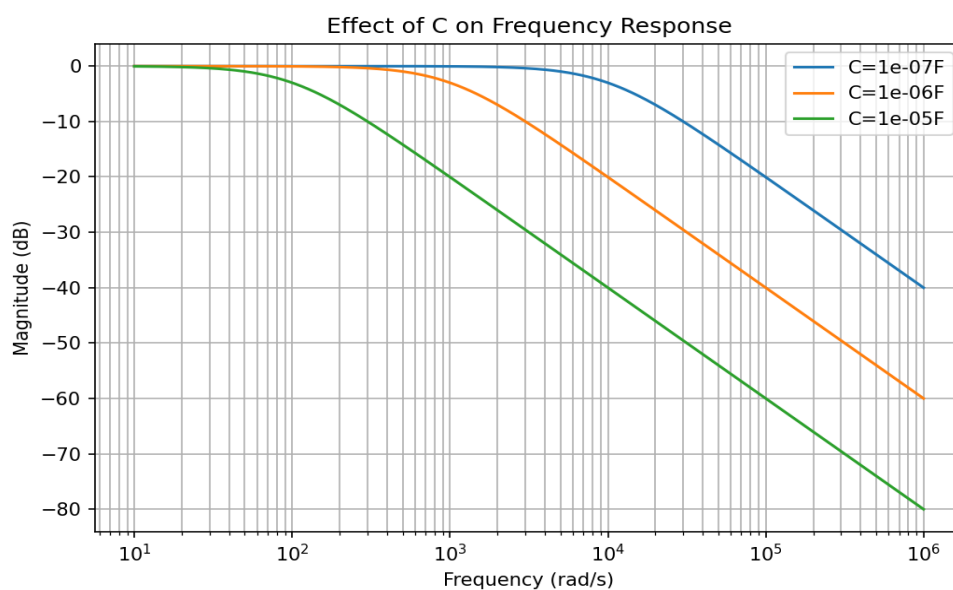


Figure 5. Effect of capacitance on the frequency response of the RC circuit.

As the capacitance increases, the cutoff frequency shifts to lower values, resulting in earlier attenuation of high-frequency signals.

Smaller capacitance values lead to a higher cutoff frequency, allowing more high-frequency components to pass. In contrast, larger capacitance values increase the system's filtering effect on high-frequency signals.

This confirms that capacitance plays a key role in determining the time constant and frequency response of the system.

Both resistance and capacitance determine the cutoff frequency through the relation $\omega_0 = 1/(RC)$, indicating that the system behavior can be tuned by adjusting these parameters.

This confirms that the RC circuit can be tuned to achieve desired filtering performance in practical applications.

Experimental Results

The theoretical model of the RC circuit was first derived analytically and then implemented in Python to obtain the simulated frequency response.

Python was used for simulation of the theoretical model, while Excel was used for processing and visualization of experimental data.

The experimental data was collected from physical measurements and compared with the simulation results.

The following are actual experimental data (only two scenarios are displayed due to limited conditions).

Table 1. Experimental data for RC circuit R = 5100Ω, C = 47nF

Experiment 1						
f [HZ]	V ₁ [V]	V ₂ [V]	V ₁₂ [V]	T(jω)	L(ω)	φ(ω)
50	1.0631	1.0595	0.0778	0.9966	-0.0295	-4.1966
180	1.0626	1.026	0.2697	0.9656	-0.3044	-14.7028
320	1.0611	0.958	0.4501	0.9028	-0.8878	-25.0984
510	1.0588	0.8442	0.6334	0.7973	-1.9674	-36.7419
660	1.0571	0.7554	0.734	0.7146	-2.9188	-43.9742
750	1.0561	0.7061	0.78	0.6686	-3.4968	-47.6076
850	1.0552	0.6557	0.8213	0.6214	-4.1326	-51.1060
950	1.0542	0.6101	0.8543	0.5787	-4.7504	-54.1297
1150	1.0525	0.532	0.928	0.5055	-5.9262	-61.7765
2100	1.0471	0.3177	0.9919	0.3034	-10.3594	-71.2842
3000	1.0434	0.2245	1.0129	0.2152	-13.3447	-76.0277
4000	1.0406	0.1683	1.0207	0.1617	-15.8240	-78.5813
5000	1.0384	0.1332	1.0234	0.1283	-17.8372	-79.8668
6000	1.037	0.1091	1.0247	0.1052	-19.5591	-80.5216
		R1 [Ω]	C1 [nF]	f1 [HZ]		
		5100	47	664		

Table 2. Experimental data for RC circuit R = 1000Ω, C = 220nF

Experiment 2						
f[HZ]	V ₁ [V]	V ₂ [V]	V ₁₂ [V]	T(jω)	L(ω)	φ(ω)
70	1.0607	1.0842	0.1003	1.0222	0.1903	-5.2115
180	1.0585	1.0571	0.2422	0.9987	-0.0115	-13.1474
270	1.0555	1.0217	0.3073	0.9680	-0.2827	-16.9133
380	1.0505	0.9663	0.4681	0.9198	-0.7257	-26.4192
580	1.0415	0.8532	0.6312	0.8192	-1.7322	-37.2717
690	1.0367	0.7919	0.6973	0.7639	-2.3397	-42.2374
730	1.0351	0.7705	0.7179	0.7444	-2.5642	-43.8804
900	1.0294	0.6863	0.7768	0.6667	-3.5214	-48.9854
1200	1.0155	0.5491	0.8533	0.5407	-5.3406	-57.1691
2000	1.05	0.3586	0.9315	0.3415	-9.3316	-61.1468
3000	0.9945	0.2428	0.9686	0.2441	-12.2471	-76.8586
4000	0.9906	0.1821	0.9677	0.1838	-14.7118	-77.5132
5000	0.9881	0.1446	0.971	0.1463	-16.6927	-79.0236
		R2[Ω]	C2[nF]	f2[HZ]		
		1000	220	724		

In this experiment, the magnitude is calculated as the ratio between output and input voltage, while the phase is obtained using the cosine theorem based on measured voltages.

The V12 measurement is used to calculate angles via the cosine theorem, thereby obtaining the test angle values using the connection method shown in the figure below.

$$\cos\theta = \frac{V_1^2 + V_2^2 - V_{12}^2}{2V_1V_2}$$

$$\theta = \arccos\left(\frac{V_1^2 + V_2^2 - V_{12}^2}{2V_1V_2}\right)$$

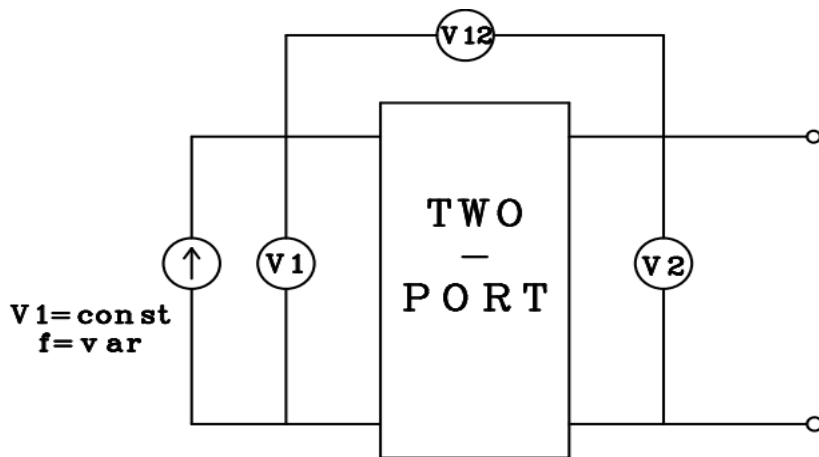


Figure 6. Link diagram

Create charts using Excel software to generate the following icons for observation and analysis.

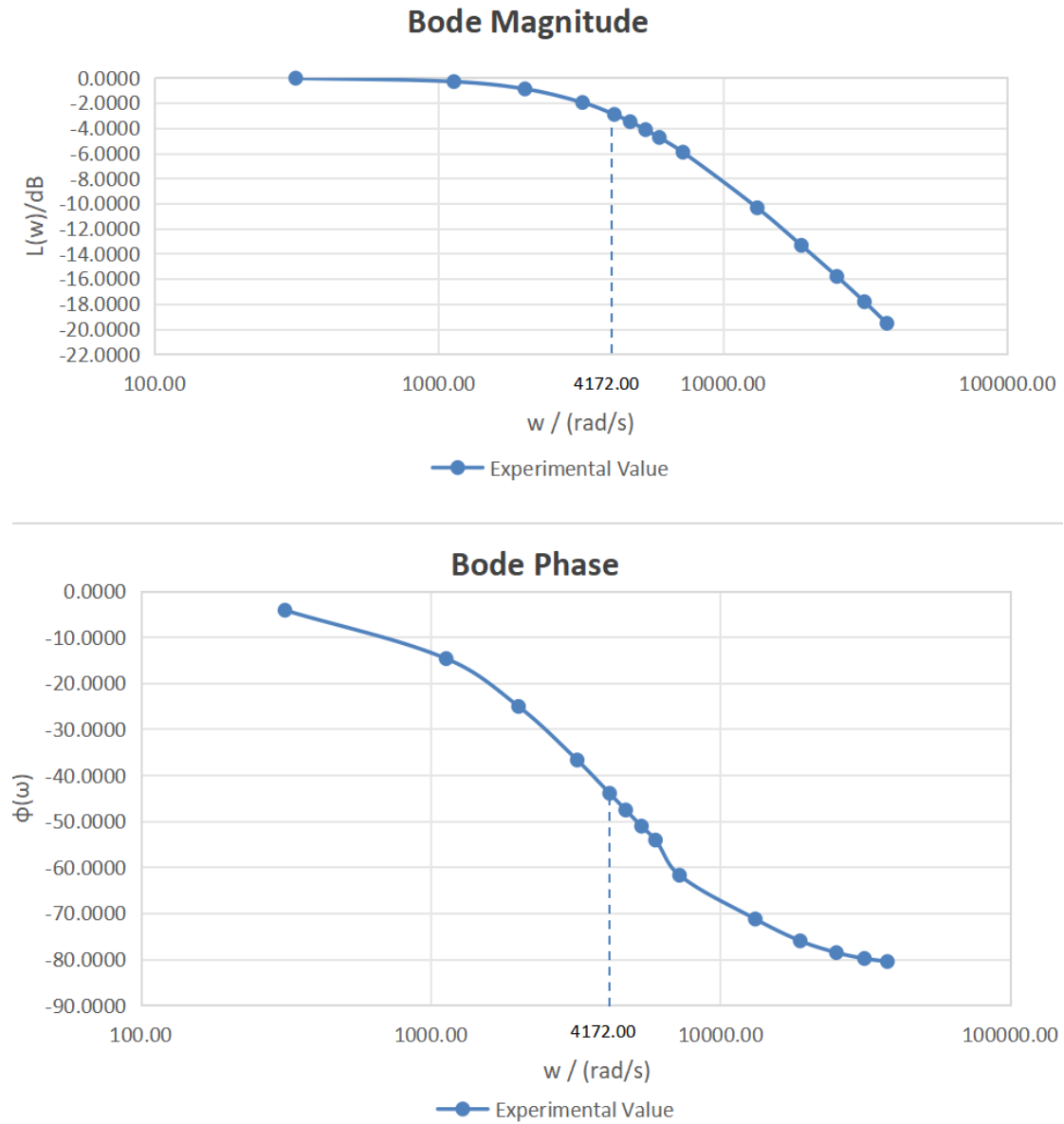


Figure 7. Comparison between simulation and experimental results for $R = 5100 \, \Omega$ and $C = 47 \, \text{nF}$

The experimental results show a typical low-pass filter behavior. The magnitude remains close to 0 dB at low frequencies and decreases as the frequency increases, indicating attenuation of high-frequency components.

The phase response gradually decreases from near 0° toward -90° , showing increasing phase delay. The cutoff frequency can be identified around $\omega \approx 4172 \, \text{rad/s}$, where the magnitude begins to drop significantly and the phase approaches -45° .

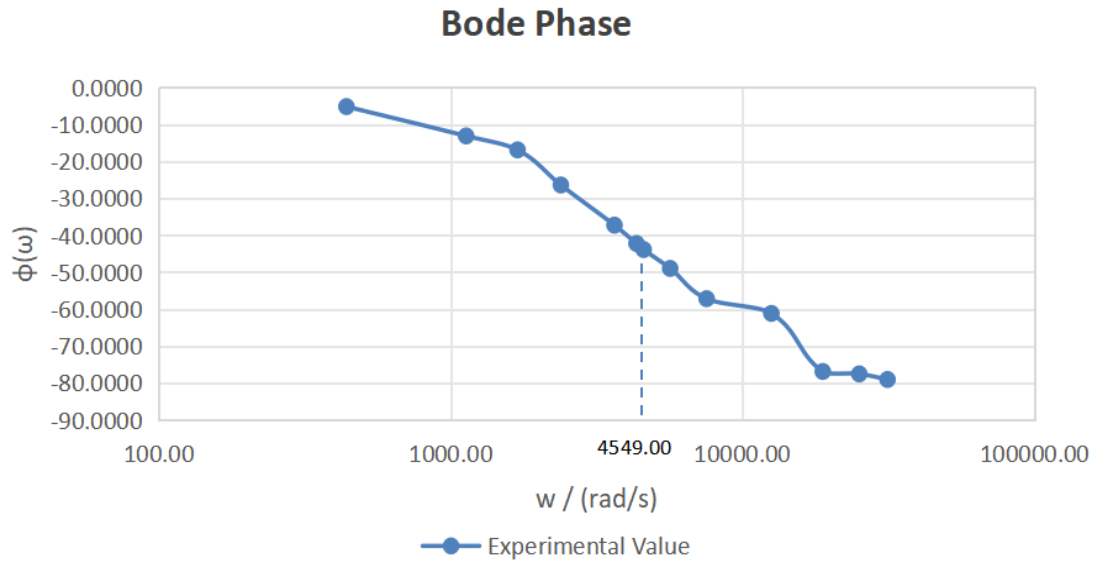
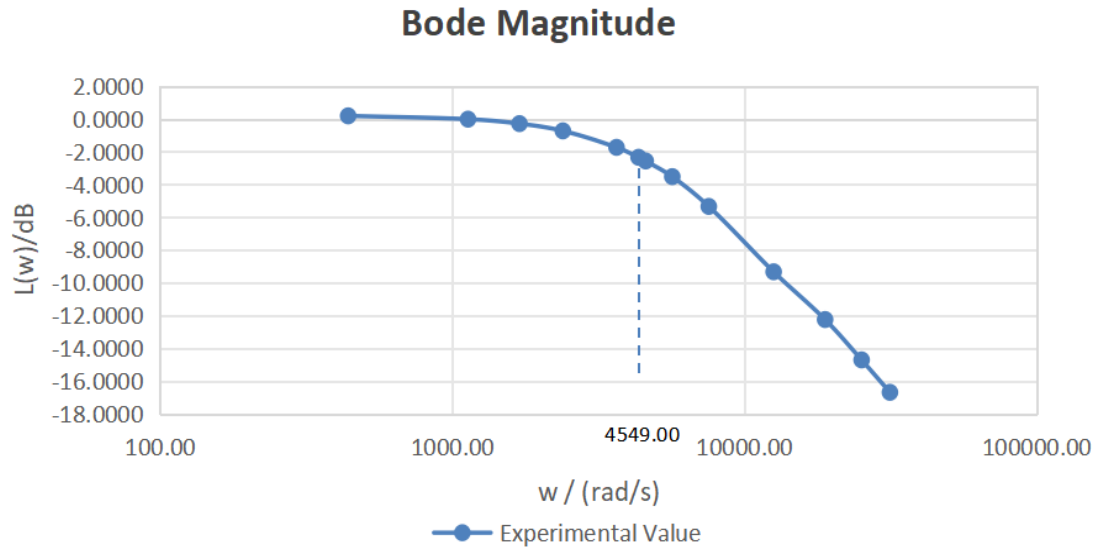


Figure 8. Comparison between simulation and experimental results for $R = 1000 \, \Omega$ and $C = 220 \, \text{nF}$

A similar low-pass behavior is observed in this configuration. The magnitude remains near 0 dB at low frequencies and decreases as frequency increases.

The phase shifts from approximately 0° to -90° , consistent with first-order system characteristics. The cutoff frequency is observed around $\omega \approx 4549 \, \text{rad/s}$, which differs from the previous configuration due to the change in resistance and capacitance values.

Comparison and Discussion

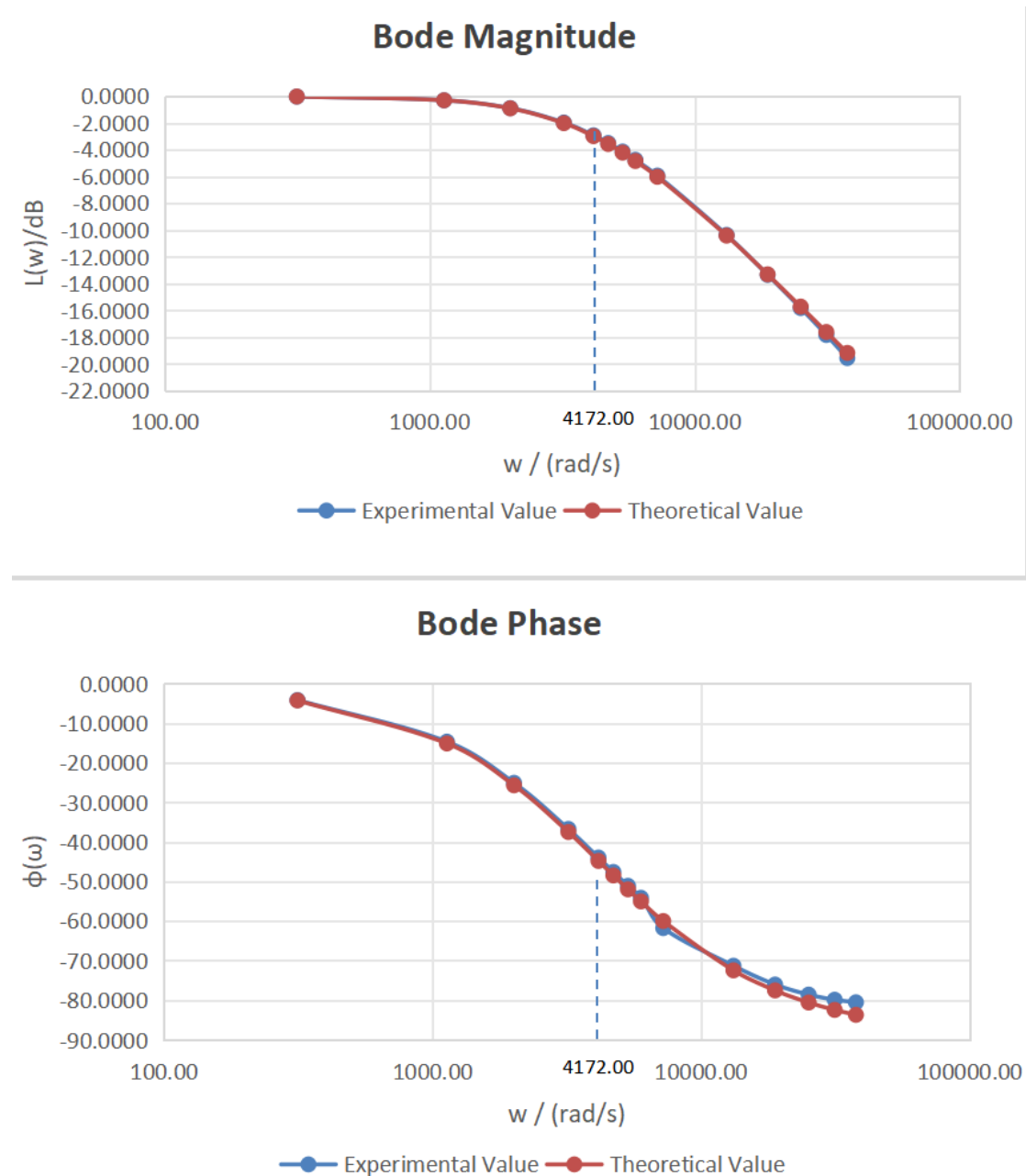


Figure 9. Comparison between simulation and experimental results $R = 5100\Omega$, $C = 47\text{nF}$

The experimental data shows strong agreement with the simulated response derived from the theoretical model. Both magnitude and phase curves follow the same trend across the frequency range.

At low frequencies, the magnitude remains close to 0 dB in both cases, while at higher frequencies, both curves exhibit similar attenuation behavior. Minor deviations at higher frequencies may be attributed to non-ideal components and measurement uncertainties.

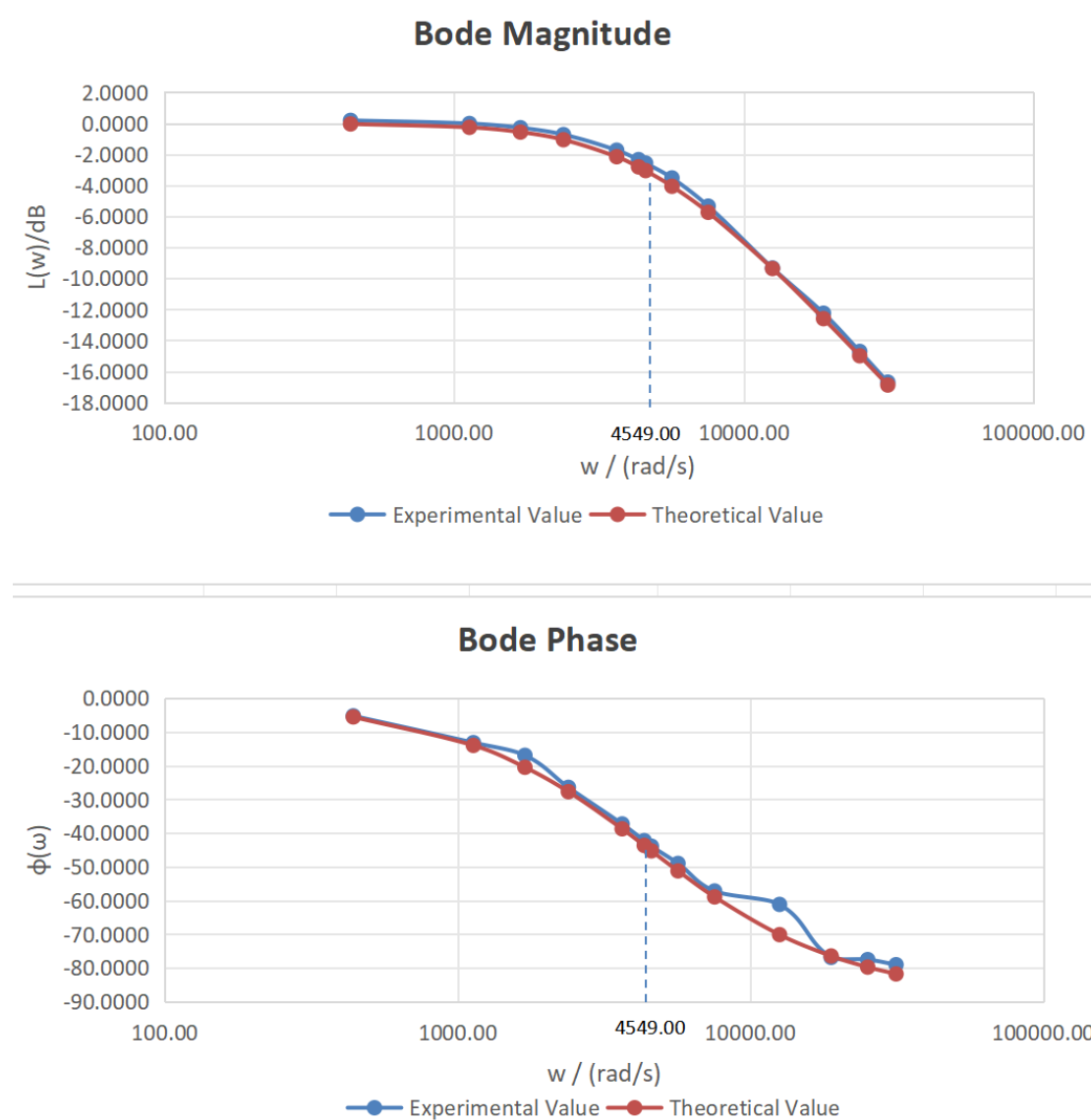


Figure 10. Comparison between simulation and experimental results $R = 1000 \, \Omega$, $C = 220\text{nF}$

The comparison demonstrates that the experimental results closely match the simulated response. The magnitude decreases with increasing frequency, and the phase shows a consistent shift toward -90° , in agreement with theoretical predictions.

Slight discrepancies between simulation and experiment can be observed, particularly at higher frequencies, which are likely caused by parasitic effects and limitations in measurement accuracy.

The experimentally observed cutoff frequencies are consistent with theoretical predictions, indicating that the model accurately captures the system characteristics. Although minor deviations appear at higher frequencies, these can be attributed to non-ideal components and measurement uncertainties.

Overall, the simulation serves as a bridge between theoretical analysis and experimental

observations, demonstrating that the system behavior can be reliably predicted.

Limitations

The analysis assumes ideal components, while in practical experiments, parasitic effects and measurement uncertainties may introduce deviations. Future work could consider more complex models including non-ideal behavior.

Additionally, component tolerances and environmental factors may also affect the accuracy of the results.

Conclusion

This project presents a comprehensive analysis of the frequency response of an RC low-pass filter through theoretical derivation, numerical simulation, and experimental validation.

The theoretical model was first established and implemented using Python to obtain the simulated frequency response. Experimental data was then collected and processed using Excel for comparison and visualization.

The results demonstrate that the experimental data closely follows the simulated response across the frequency range. Key characteristics such as low-pass behavior and cutoff frequency are consistent with theoretical predictions, confirming the accuracy of the model.

Although minor discrepancies are observed at higher frequencies, they can be reasonably explained by non-ideal components and measurement limitations.

Overall, this project illustrates the effectiveness of combining analytical modeling, simulation, and experimental verification to understand and predict system behavior in practical engineering applications.